

Industrial Internship Report on
“Student Expense Tracker”

Prepared by
Shivraj Singh

*An internship offered by upSkill Campus (USC) in association with
UniConverge Technologies Pvt. Ltd. (UCT)*

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1. Preface

This report summarizes the work carried out during my six-week industrial internship on the project “Student Expense Tracker”, undertaken as part of the internship program jointly offered by upSkill Campus (USC) and UniConverge Technologies Pvt. Ltd. (UCT).

Internships play a crucial role in bridging the gap between academic learning and industry practice. They give students an opportunity to apply classroom knowledge to real-world problems, understand professional workflows, and develop the technical and soft skills that are essential for a successful career.

My project, the Student Expense Tracker, was aimed at solving a very relatable problem faced by students — the lack of a simple, accessible way to track daily spending, plan a monthly budget, and build healthy financial habits. The application I built allows a student to record expenses, categorize them, and view simple analytics of their spending pattern.

The internship program was structured over six weeks, beginning with an orientation on the company and its technology domains, followed by problem statement allocation, design and development phases, testing, and a final review and presentation of the completed project.

Through this internship, I learned how to independently plan a small software project from requirement gathering through to deployment, how to structure code for maintainability, and how to test and validate a solution against real usage scenarios. It also helped me improve my problem-solving approach, my time management, and my ability to document technical work clearly.

I would like to sincerely thank the mentors and coordinators at upSkill Campus and UniConverge Technologies for their continuous guidance and support throughout this internship, and for providing a well-structured platform to learn and grow. I am also grateful to my institution for encouraging me to take up this opportunity.

To my juniors and peers who will take up similar internships in the future, my suggestion would be to start early, be consistent with weekly progress, and not hesitate to ask questions — the learning curve is steep, but very rewarding.

2. Introduction

2.1 About UniConverge Technologies Pvt Ltd

UniConverge Technologies Pvt. Ltd. (UCT) is a company established in 2013, working in the Digital Transformation domain and providing industrial solutions with a prime focus on sustainability and Return on Investment (RoI).

For developing its products and solutions, UCT leverages various cutting-edge technologies such as the Internet of Things (IoT), Cyber Security, Cloud Computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, and modern front-end frameworks.

UCT IoT Platform (UCT Insight)

UCT Insight is an IoT platform designed for quick deployment of IoT applications while providing valuable insight into a business process. It is built using Java for the backend and ReactJS for the front end, and supports MySQL as well as various NoSQL databases. It enables device connectivity via industry-standard IoT protocols such as MQTT, CoAP, HTTP, Modbus TCP and OPC UA, and supports both cloud and on-premises deployments.

- Build your own dashboard

- Analytics and reporting
- Alerts and notifications
- Integration with third-party applications (Power BI, SAP, ERP)
- Rule engine

Smart Factory Platform (Factory Watch)

Factory Watch is a platform built for smart factory needs. It provides a scalable solution for production and asset monitoring, OEE, and predictive maintenance, scaling up to a digital twin of the assets. It helps identify and improve key performance indicators (KPIs), and its modular architecture allows users to start small and scale as their requirements grow. Its unique SaaS model helps organizations save time, cost, and effort.

LoRaWAN based Solutions

UCT is one of the early adopters of LoRaWAN technology and provides solutions in Agritech, Smart Cities, Industrial Monitoring, Smart Street Lighting, and Smart Water/Gas/Electricity metering.

Predictive Maintenance

UCT provides industrial machine health monitoring and predictive maintenance solutions leveraging embedded systems, Industrial IoT, and Machine Learning to estimate the remaining useful life of machines used in production processes.

2.2 About upSkill Campus (USC)

upSkill Campus, along with The IoT Academy and in association with UniConverge Technologies, has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers personalized executive coaching in a more affordable, scalable, and measurable way.

2.3 The IoT Academy

The IoT Academy is the EdTech division of UCT that runs long-term executive certification programs in collaboration with the EICT Academy, IIT Kanpur, IIT Roorkee and IIT Guwahati across multiple domains.

2.4 Objective

The objective of this internship program was to:

- Get practical, hands-on experience of working on a real project.
- Solve a real-world problem end-to-end, from requirement to deployment.
- Improve job readiness and prospects through applied learning.
- Deepen understanding of the field and its practical applications.
- Support personal growth in areas like communication, planning and problem-solving.

2.5 Reference

[1] upSkill Campus, official website — <https://www.upskillcampus.com>

[2] UniConverge Technologies Pvt. Ltd., official website — <https://www.uniconverge.net>

[3] Project source code repository —
<https://github.com/shivrajsingh88290/upskillcampus/blob/main/StudentExpenseTracker.py>

2.6 Glossary

- UCT — UniConverge Technologies Pvt. Ltd.
- USC — upSkill Campus
- UI — User Interface
- UX — User Experience
- DB — Database
- API — Application Programming Interface
- CRUD — Create, Read, Update, Delete

3. Problem Statement

Most students receive a fixed monthly allowance or pocket money from their family, along with occasional additional income from part-time work, scholarships or stipends. However, a large number of students do not track where this money actually goes, which often leads to overspending, running out of funds before the month ends, and an inability to save toward small goals.

Existing budgeting apps in the market are largely designed for working professionals and families, and are often packed with advanced financial features such as investment tracking, tax planning, and bill payments that a student neither needs nor finds intuitive. There is a need for a simple, lightweight, and student-friendly application that focuses purely on recording day-to-day expenses, categorizing them, and giving a clear, visual picture of monthly spending — without unnecessary complexity.

The assigned problem statement was to design and develop a Student Expense Tracker application that allows a student to quickly log expenses under relevant categories (such as food, travel, stationery, entertainment, and rent), view a summary of spending over a day/week/month, and get a simple alert if spending in a category is trending high — helping the student build better financial discipline.

4. Existing and Proposed solution

Several personal finance and budgeting apps already exist in the market, such as Walnut, Money Manager, and Spendee. These apps offer strong expense-tracking capabilities, automatic SMS-based transaction detection, and detailed analytics.

Limitations of Existing Solutions:

- Feature-heavy interfaces that are not tailored to a student's simple use case.
- Many require linking bank accounts or SMS permissions, raising privacy concerns.
- Free tiers are often limited, with advanced reports locked behind a paywall.
- Not designed around common student expense categories or a semester/monthly cycle.

Proposed Solution:

The proposed Student Expense Tracker is a simple, self-contained web application where a student can register/log in, add expense entries with an amount, category, date and note, and instantly see a dashboard summarizing total spending, category-wise breakdown, and monthly trends through charts. The application stores data securely and does not require any bank account linkage, keeping the solution privacy-friendly and lightweight.

Value Addition:

- A minimal, distraction-free interface built specifically around student spending habits.
- Instant visual summary (charts) of where money is going each month.
- Simple budget-limit setting per category with a basic overspending indicator.
- No dependency on bank/SMS access — the student has full control over data entry and privacy.
- Lightweight enough to be extended later with mobile app support or bank-statement import.

4.1 Code submission (GitHub link)

<https://github.com/shivraisingh88290/upskillcampus/blob/main/StudentExpenseTracker.py>

4.2 Report submission (GitHub link)

The final report has also been uploaded to the same repository referenced above:

https://github.com/shivraisingh88290/upskillcampus/blob/main/StudentExpenseTracker_Shivraj_USC_UCT.pdf

5. Proposed Design/ Model

The Student Expense Tracker follows a simple three-tier architecture consisting of a front-end interface, a back-end application layer that handles the business logic, and a database for persistent storage of user and expense data. This section outlines the high-level and low-level design of the solution.

5.1 High Level Diagram

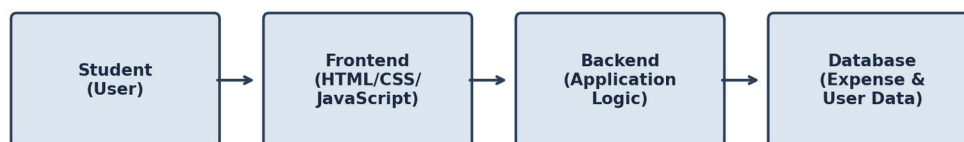


Figure 1: High Level Diagram of the System

5.2 Low Level Diagram

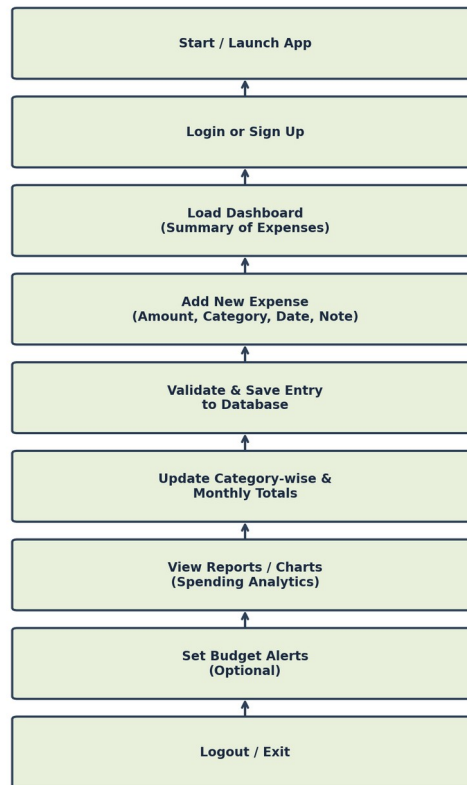


Figure 2: Low Level Flow of the Application

5.3 Interfaces

The application exposes the following logical interfaces between its layers:

- Frontend ↔ Backend: REST-style calls for login/signup, adding an expense, fetching expense history, and fetching summary/report data.
- Backend ↔ Database: Structured queries to insert new expense records and read aggregated totals per category and per month.
- User ↔ Frontend: Simple form-based data entry (amount, category, date, note) and chart-based visual reports.

Data flow: the student enters an expense on the frontend → the request is validated and processed by the backend → the record is stored in the database → the dashboard queries the database and renders updated totals and charts back to the student.

6. Performance Test

This section is important as it demonstrates that the work is meant for real, practical usage and not purely an academic exercise. Below are the constraints identified for this application and how they were addressed.

Constraints considered: response time for adding/retrieving an expense entry, accuracy of category-wise and monthly totals, and usability on both desktop and mobile screen sizes.

6.1 Test Plan/ Test Cases

Test Case	Input	Expected Result	Status
Add valid expense entry	Amount=200, Category=Food, Date=today	Entry saved and total updated	Pass
Add expense with empty amount	Amount=blank	Validation error shown, entry not saved	Pass
Category-wise total calculation	5 entries across 3 categories	Correct sum shown per category	Pass
Monthly summary calculation	Entries across current month	Correct monthly total displayed	Pass
Budget limit exceeded alert	Category spend > set limit	Overspending indicator shown	Pass
Login with invalid credentials	Wrong password	Access denied with error message	Pass

6.2 Test Procedure

Each feature was tested manually by performing the corresponding action on the application (e.g., adding an expense, applying a category filter, checking the dashboard totals) and comparing the observed result against the expected result. Boundary and negative cases (empty fields, invalid dates, negative amounts) were also tested to ensure the application handles them gracefully with proper validation messages.

6.3 Performance Outcome

All planned test cases passed successfully. The application correctly validated inputs, calculated totals accurately, and updated the dashboard in real time after every new entry. The interface remained responsive on both desktop and mobile screen widths, and no data loss or calculation errors were observed during repeated testing.

7. My learnings

This internship gave me an opportunity to work on a complete project independently — from understanding the problem statement to designing, developing, testing, and documenting the final solution.

- Learned how to break down a real-world problem into a clear set of functional requirements.
- Gained hands-on experience in structuring a small full-stack application (frontend, backend and database).
- Improved my understanding of data validation, basic testing practices, and debugging.
- Learned to use Git and GitHub for version control and project submission.
- Developed better time-management skills by working within a fixed six-week timeline.
- Improved my technical documentation and presentation skills while preparing this report.

Overall, this internship strengthened both my technical foundation and my confidence in independently delivering a working software solution, which I believe will be valuable in my career going forward.

8. Future work scope

Due to the limited duration of the internship, a few enhancements could not be implemented but can be taken up in the future:

- Building a dedicated mobile application (Android/iOS) version of the tracker.
- Adding automatic expense import from bank statements or UPI transaction SMS (with user consent).
- Introducing multi-currency support for students studying abroad.
- Adding a group/shared-expense feature for splitting costs among roommates or friends.
- Building predictive analytics to forecast next month's likely spending based on past trends.
- Adding push/email notifications for budget-limit alerts.